Gear Transmission Design and Multibody Modeling for a Ball-and-Beam System

Analytical evaluation, simulation architecture, and documented results

Abstract

This report examines gear transmission alternatives for actuating a ball-and-beam mechanism. The available project comprises an externally meshing spur-gear model, a controlled spur-gear ball-and-beam model, a worm-driven variant, supporting geometry, and two recorded gear animations. A file-based evaluation is combined with analytical calculations to characterize transmission ratios, dimensional consistency, actuation assumptions, and feedback implementation. The 20/80-tooth spur pair provides a geometric reduction of 4:1, whereas the 10/50-tooth pair provides 5:1. The saved worm-constraint dimensions imply an ideal reduction of approximately 25.63:1. At an illustrative input speed of 30 rpm, the corresponding calculated output speeds are 7.50, 6.00, and 1.17 rpm. The recorded animations document geometric motion but do not provide calibrated speed or torque measurements. The controlled models contain two PID blocks; however, their secondary feedback paths require dimensional review because translational sensing is followed by an angular conversion. The study therefore establishes the implemented modeling structure and analytical performance while identifying the validation required before quantitative closed-loop performance can be established.

Keywords: ball-and-beam system; spur gears; worm transmission; multibody dynamics; PID control

Study objective and scope

The objective is to relate the supplied transmission geometry and model settings to the mechanical requirements of beam actuation. The evaluation covers the saved model structure, analytical kinematics, available visual evidence, and reproducibility. No new dynamic simulation or physical experiment is represented as an observed result.

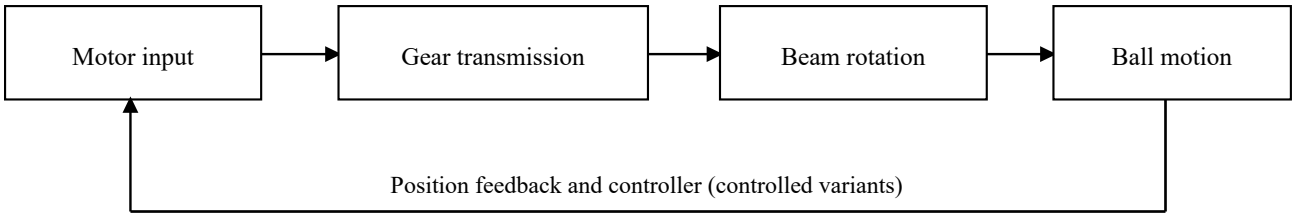


Figure 1. Functional organization of the intended controlled mechanism. This schematic describes the design objective; the sensing and unit definitions in the saved implementations require the checks discussed in Section 4.

1. Mechanical and Analytical Framework

1.1 Transmission requirements

Tilting the beam changes the gravitational component acting along its surface and thereby changes ball motion. A reduction stage can increase the available output torque and reduce output angular speed. The appropriate reduction depends on beam inertia, the permitted angular range, actuator travel, and the response speed required by the controller. Transmission ratio alone does not establish positioning accuracy; backlash, compliance, friction, sensing, and actuator resolution also affect performance.

1.2 Spur-gear kinematics and torque

For an external spur pair with a common module, the pitch diameter d, pitch radius r, reduction i, and center distance a follow from the tooth counts z. The negative speed sign represents opposite directions of rotation. The torque relation assumes steady transmission and an efficiency η; the ideal calculations in this report set $\eta = 1$.

$$d = m\,z, \quad r = m\,z / 2, \quad i = z_2 / z_1 = r_2 / r_1$$

$$a = r_1 + r_2, \quad \omega_2 = -\omega_1 / i, \quad T_2 = \eta\, i\, T_1$$

The saved common-gear blocks specify *pitch-circle radii*. Consequently, the variables r1 and r2 determine the active reduction. A stored GearRatio field equal to 1 is inactive under that specification and should not be interpreted as evidence of a unity transmission ratio.

1.3 Beam inertia and actuator loading

For the nominal analytical beam mass $M = 0.60$ kg and length $L = 0.30$ m, the inertia depends on the location of the rotation axis. A uniform beam rotating about its center has one-quarter of the inertia of the same beam rotating about an end.

$$I_{center} = M\,L^2 / 12 = 0.00450 \text{ kg·m}^2$$

$$I_{end} = M\,L^2 / 3 = 0.0180 \text{ kg·m}^2$$

Using the nominal torque value $T = 0.92$ N·m as an illustrative upper-bound input gives $\alpha = T/I = 204.4$ rad/s² for a central axis and 51.1 rad/s² for an end axis. These are ideal rigid-body estimates that omit gearing, ball loading, friction, and torque-speed behavior. Stall torque is not a continuous operating torque.

1.4 Reduced ball dynamics

For a ball rolling without slip on a slowly tilting beam, a reduced relation is $(m_b + J_b/R^2)\ddot{x} = m_b g \sin\theta$. A homogeneous solid sphere gives $\ddot{x} \approx (5/7)g\theta$ at small angles. This approximation neglects beam angular-rate effects and contact transients. The supplied contact models allow slip, so this reduced relation is explanatory rather than a verified equivalent of the full model.

2. Spur-Driven Model Configurations

2.1 Baseline beam mechanism

The baseline model connects two CAD gear solids through a common-gear constraint and revolute joints. A linkage assembled from rigid transforms, brick solids, and revolute subsystems transfers output rotation to the beam. The effective initialization values are m = 0.2 cm, $z_1 = 20$, $z_2 = 80$, $r_1 = 2$ cm, and $r_2 = 8$ cm. The resulting center distance is 10 cm. The small input gear receives prescribed joint motion from a sine signal, rather than torque from a detailed motor model.

Quantity	Baseline spur model	Controlled spur model
Gear teeth	20 / 80	10 / 50
Geometric module	2 mm	3 mm
Pitch radii	20 / 80 mm	15 / 75 mm
Required center distance	100 mm	90 mm
CAD gear thickness	5 mm	3.5 mm
Beam dimensions	400 × 30 × 10 mm	500 × 30 × 10 mm
Ball representation	No ball solid in this model	Sphere, radius 10 mm
Simulation stop time	20 s	10 s

Table 1. Geometry and saved configuration of the two spur-driven models. Required pitch radii and center distances are calculated from the corresponding CAD tooth specifications.

2.2 Controlled spur variant

The second spur model adds a spherical body, a six-degree-of-freedom joint, spatial contact, sensing, and two PID blocks. The geometry is a distinct 10/50-tooth set and therefore requires its own pitch-radius initialization. Applying the baseline 20/80-tooth initialization unchanged would impose a 4:1 constraint on geometry whose tooth ratio is 5:1.

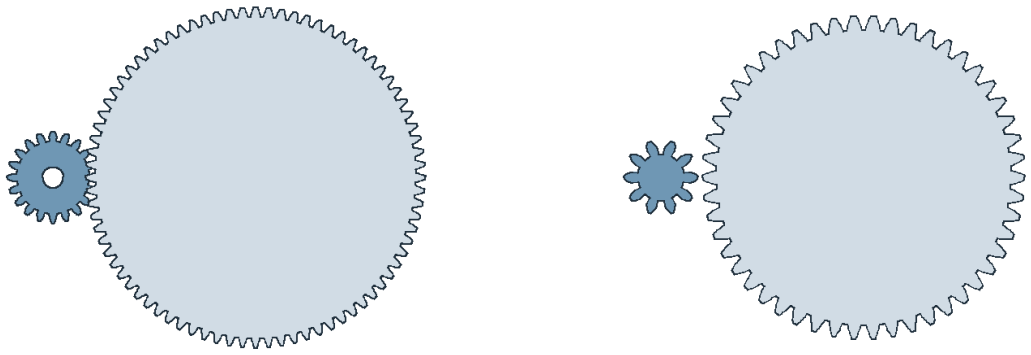


Figure 2. Orthographic silhouettes reconstructed from the supplied STL triangle vertices. Left: the 20/80-tooth pair. Right: the 10/50-tooth pair. Gear sizes are shown on a common scale; the horizontal placement is illustrative and does not depict assembled center distances.

The baseline and controlled beam dimensions differ from the nominal 300 mm analytical beam. Consequently, the nominal inertia calculation in Section 1 cannot be assigned directly to either saved multibody assembly.

3. Worm-Driven Multibody Model

The worm-driven variant combines a STEP worm solid, a helical STL gear solid, a parameterized worm-gear constraint assembly, a beam linkage, and a spherical ball with spatial contact. The assembly uses right-handed worm kinematics and includes viscous joint terms. The CAD geometry provides the visual and inertial representation, while the constraint parameters prescribe the kinematic relationship.

Saved parameter	Value
Worm pitch radius r_w	8.5 mm
Gear pitch radius r_g	16 mm
Worm lead angle λ	4.2°
Handedness	Right-hand
Worm / gear viscous terms	0.01 / 0.01 N·m/(rad/s)
Main beam dimensions	9 × 120 × 2 mm
Ball radius and density	3 mm; 6,000 kg/m³
Gravity vector	[0, -9.8, 0] m/s²
Main / archived stop time	30 s / 10 s

Table 2. Saved worm-driven model parameters.

Equating the worm axial advance per revolution to gear pitch-circle travel gives the following ideal speed ratio for the saved constraint geometry:

$$i_w = r_g / (r_w \tan \lambda) = 25.63$$

This value is derived from the radii and lead angle, rather than an assumed tooth-count-to-start-count ratio. The supplied STL name identifies 15 teeth, but an integer worm start count is not established by the saved parameter set. Geometric compatibility between the CAD surfaces and the ideal constraint should therefore be checked separately.

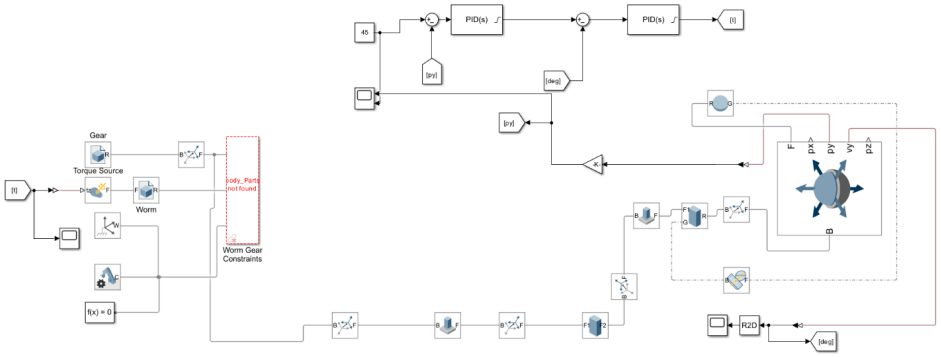


Figure 3. Worm-driven control and multibody layout documented in the project. The figure records the model arrangement; it is not a quantitative response plot.

The main worm model uses a position command of +45, while the archived variant uses -10. The position gain of 1,000 indicates a millimeter numerical scale. Self-locking cannot be inferred from this kinematic constraint and viscous terms alone; it requires a suitable friction model and a back-driving evaluation.

4. Feedback and PID Implementation

Both controlled variants contain two continuous-time PID blocks connected in a cascade-like structure. The first compares a scaled ball-position signal with a constant command. Its output becomes the command to the second block. The spur variant applies the resulting signal through a Simulink-PS converter to a motion-actuated revolute joint. The worm variant routes the second controller output to an external torque source.

Variant / block	K_p	K_i	K_d	N	Output limits
Spur / first	1.49186	0.82954	0.25215	18.4653	±30
Spur / second	0.46179	0.55542	-0.02185	20.9570	±3
Worm / first	0.006647	0.003505	0.002183	26.2402	±π/6
Worm / second	2.21543	0.17190	3.27889	21.7131	±1.5

Table 3. Saved PID coefficients and numerical saturation limits. Coefficient units depend on each signal definition. N is the derivative-filter coefficient in the continuous PID form.

$$C(s) = K_p + K_i/s + K_d N\, s/(s + N)$$

4.1 Signal scaling and secondary feedback

The controlled spur model scales ball displacement by 100, consistent with converting meters to centimeters, and uses a position command of -15. Its six-degree-of-freedom joint enables x-position and x-velocity sensing. The second sensed signal passes through a Radians to Degrees block before entering the second error calculation. This is a translational sensing path, rather than a revolute-angle measurement.

In the worm model, the first converter receives the y-position sensing path and a gain of 1,000 scales displacement. The next sensing path, associated with enabled y-velocity sensing, likewise passes through a Radians to Degrees block. Multiplication by 180/π does not convert linear velocity into a beam angle. A physically meaningful inner loop requires either an angular sensor with consistent units or an explicitly formulated velocity-control interpretation.

4.2 Saturation and actuation interpretation

Output limiting is enabled in all four controllers, while the saved anti-windup mode is none. The saturation values cannot be compared physically until the controller input and output units are established. The negative derivative coefficient in the second spur PID also requires review of loop sign conventions and closed-loop behavior; its value alone does not establish stability or instability.

The motion-actuated spur joint computes the torque needed to impose its input motion. It does not reproduce a servo torque limit or torque-speed characteristic. The worm torque-source model is closer to torque actuation, but no detailed electrical motor, supply, or internal servo dynamics are present. These distinctions limit direct comparison of actuator demand between the two variants.

5. Calculated Transmission Results

Table 4 compares the two supplied spur geometries and the saved worm-constraint dimensions under common illustrative inputs. The input speed is 30 rpm and the input torque is 0.92 N·m. These values define a calculation scenario; simultaneous delivery of the quoted torque and speed has not been measured or established for the actuator.

Transmission	Reduction i	Output speed	Ideal output torque
Spur, 20 / 80 teeth	4.00	7.50 rpm	3.68 N·m
Spur, 10 / 50 teeth	5.00	6.00 rpm	4.60 N·m
Worm constraint	25.63	1.17 rpm	23.58 N·m

Table 4. Analytical predictions at identical hypothetical input conditions. Torques assume unity efficiency and impose no gear-strength or motor operating limits.

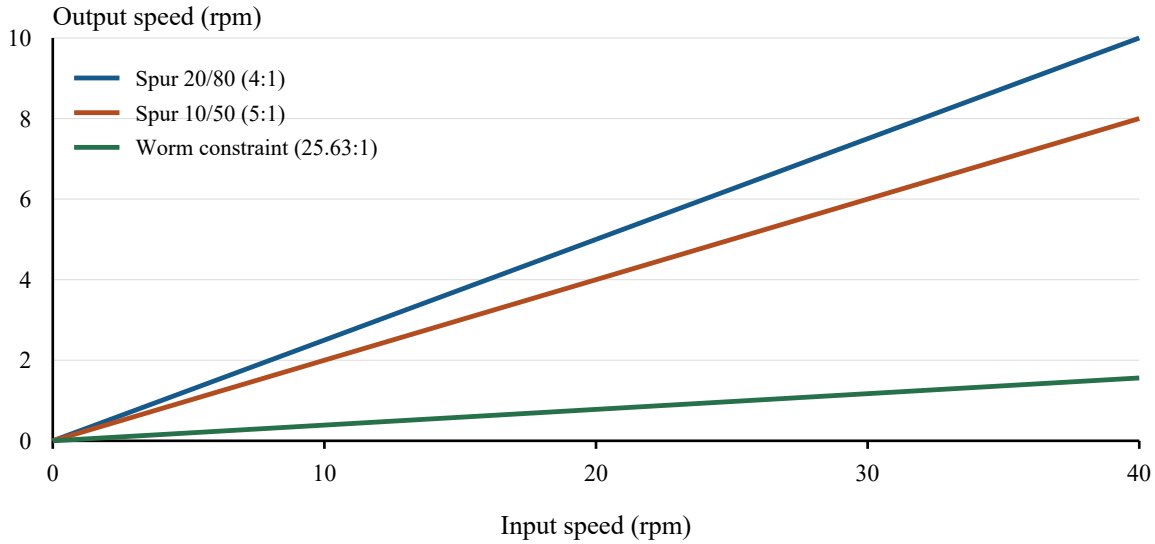


Figure 4. Calculated output-speed magnitudes as a function of input speed. Each line follows $n_{out} = n_{in}/i$ and represents ideal kinematics, not a measured or simulated response history.

5.1 Interpretation of the calculated trade-off

Changing from the 4:1 spur pair to the 5:1 pair reduces output speed by 20% and increases ideal output torque by 25% for identical input conditions. The worm constraint gives approximately 6.41 times the reduction of the 4:1 pair. Its lower output speed may be useful for fine beam motion, but greater reduction also demands more motor travel and can reduce available correction speed.

The ideal torque values do not establish practical load capacity. Efficiency, tooth strength, bearing loads, actuator limits, and transient dynamics are not included in this calculation. In particular, a larger reduction does not create additional mechanical power: $T_{out}\omega_{out} = \eta T_{in}\omega_{in}$.

5.2 Ball and beam quantities derived from saved solids

Using the saved dimensions and uniform densities, the baseline 400 mm beam has a mass of 0.120 kg, while the 500 mm controlled spur beam has a mass of 0.150 kg. The 10 mm-radius spur ball has a calculated mass of 8.38 g. The worm-model beam has a mass of 15.55 g and its 3 mm-radius ball has a mass of 0.679 g. These are geometry-based calculations, not independent mass measurements.

6. Recorded and Geometric Results

Two short videos provide the available direct evidence of gear motion. The first recording lasts approximately 13.33 s and depicts an externally meshing 20/80-tooth pair. The labels identify a common diametral pitch of 20 and a pressure angle of 20°. The second recording lasts approximately 14.87 s and depicts a compound arrangement with an internal ring and multiple external gears. Both files are encoded at 30 frames per second.

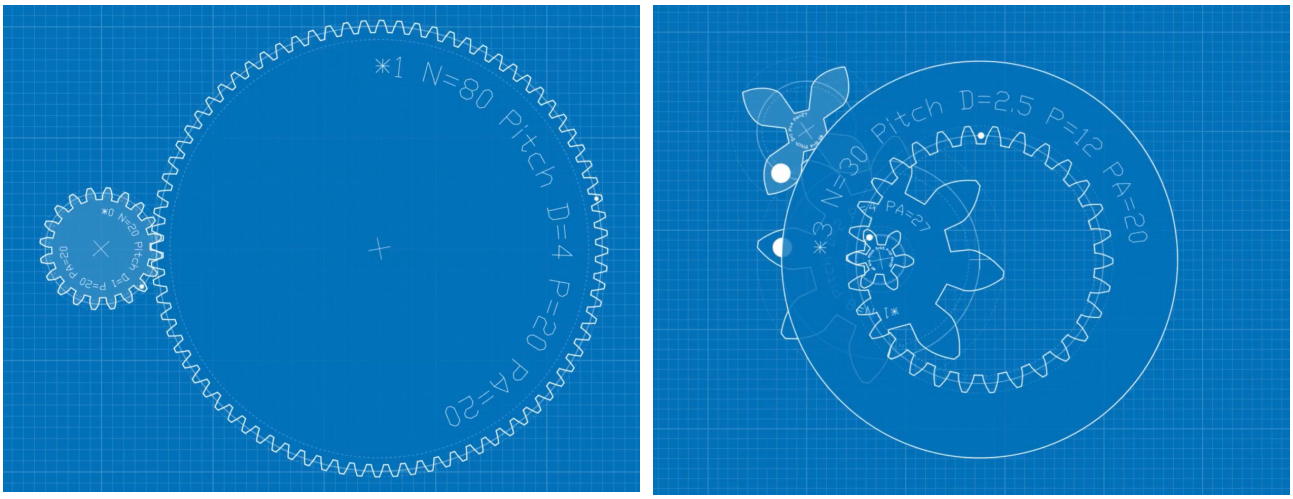


Figure 5. Frames extracted from the supplied gear animations. Left: externally meshing 20/80-tooth pair. Right: compound arrangement containing a 30-tooth internal ring and smaller gears. Color is retained only within the figures.

6.1 What the recordings establish

The recordings document visible tooth engagement and relative motion in the depicted arrangements. The first animation supports the intended 20/80 tooth-count relationship. The second documents the presence of an internal mesh and smaller external elements. Camera zooming and the absence of calibrated time-series measurements prevent these recordings from establishing numerical efficiency, torque, angular accuracy, or controller performance.

6.2 Separation of animation and CAD parameter sets

The first animation specifies diametral pitch P = 20, whereas the corresponding project STL names specify module m = 2 mm. These are distinct size specifications: if P is expressed in teeth per inch, the equivalent module is 25.4/P = 1.27 mm. The 4:1 tooth ratio is common to both, but dimensional identity between the animation and STL geometry has not been established.

The compound animation includes a small gear labeled N = 4 with P = 4, while other visible elements use P = 12. Gears in a common ideal involute mesh require compatible pitch definitions. The recording therefore requires a separate geometry audit before its labeled gears can be treated as a physically compatible transmission. A complete compound speed ratio also requires identification of rigidly shared shafts, fixed elements, and any moving carrier.

6.3 Scope of the available results

The project does not contain exported numerical records of ball position, beam angle, control torque, or tracking error. Consequently, settling time, overshoot, steady-state error, disturbance rejection, and stability margins are not reported as measured outcomes. Saved scope display limits and controller coefficients are model settings and do not substitute for response data.

7. Numerical Modeling and Verification

7.1 Contact and integration settings

Both controlled models use a spatial contact force with normal stiffness 10^6 N/m, normal damping 10^3 N·s/m, static friction coefficient 0.5, and dynamic friction coefficient 0.3. These values define a penalty-based compliant contact model with friction. They do not impose exact rolling without slip. The six-degree-of-freedom ball representation allows motion and contact behavior to emerge from the dynamics.

Model	Saved integration configuration
Baseline spur	Automatic variable-step selection; stop time 20 s
Controlled spur	Automatic variable-step selection; stop time 10 s
Main worm variant	ode15s; maximum step 0.01 s; stop time 30 s
Archived worm variant	ode15s; maximum step 0.01 s; stop time 10 s

Table 5. Saved integration settings for the principal model variants.

A maximum step size of 0.01 s is an integration limit, not a fixed controller sample period. Because compliant contact may introduce fast dynamics, reliable results require a convergence check using smaller maximum steps and stricter tolerances. Contact penetration and tangential slip should be examined alongside the ball-position trajectory.

7.2 Verification findings

Finding	Consequence for interpretation
Different beam dimensions and densities	The nominal analytical load is not identical to the multibody loads.
Different actuation approaches	Prescribed motion and applied torque do not impose comparable motor limitations.
Translational signal with angular conversion	The inner-loop physical quantity and units require correction or explicit reinterpretation.
Inactive ratio field in spur constraints	The active pitch radii must match the selected tooth geometry.
Absolute CAD paths in saved models	The original author-specific locations must be replaced on another computer.
No exported response data	Quantitative closed-loop performance remains unverified.

Table 6. Main verification findings and their implications.

7.3 Reproducible simulation procedure

For each model, select the matching CAD geometry, define all workspace variables, resolve the supporting library, and replace absolute geometry locations with project-relative paths. Verify frame orientation and gear-center placement before integration. Then check the sensor quantity, numerical scaling, feedback sign, and converter units. Only after these checks should response histories be exported for comparative evaluation.

8. Discussion and Conclusions

8.1 Transmission selection

The external spur arrangements offer a direct relationship between tooth count, center distance, and ideal speed reduction. The 4:1 pair provides the higher calculated output speed, whereas the 5:1 pair provides moderately greater ideal torque multiplication. Both retain reversibility in their ideal gear constraints, which is compatible with bidirectional beam corrections. Their practical suitability depends on backlash, bearing support, alignment, and the actuator response.

The worm-constraint arrangement gives substantially greater ideal reduction and changes the shaft orientation. This can support compact placement and small output-angle increments. However, the model does not establish self-locking, efficiency, or practical output-torque capacity. Greater reduction can also restrict beam travel when driven by a motor with limited angular excursion. For an ideal direct angular transmission, the required motor angle scales approximately as i times the output angle; the linkage further changes this relationship.

8.2 Evaluation needed for a defensible comparison

A controlled comparison should use an identical beam, ball, actuator limits, and position command across the candidate transmissions. The controller should operate on explicitly defined physical quantities. A suitable intended architecture is an outer ball-position loop commanding beam angle and an inner loop using measured beam angle or an actuator variable. The saved translational feedback conversions must be resolved before treating the existing blocks as such an architecture.

The minimum numerical outputs are ball position $x(t)$, beam angle $\theta(t)$, actuator torque $T(t)$, and tracking error $e(t)$. A position-step test can then quantify rise time, settling time, overshoot, and steady-state error. Torque saturation duration, ball slip, and loss of contact should also be examined. A disturbance test and parameter variation in friction, inertia, and gear backlash would assess whether the response remains acceptable beyond one nominal condition.

8.3 Conclusions

The supplied files establish three principal multibody configurations and two gear animations. The 20/80-tooth and 10/50-tooth spur geometries yield analytical reductions of 4:1 and 5:1, respectively. The saved worm constraint implies approximately 25.63:1. At a hypothetical 30 rpm input, their predicted output-speed magnitudes are 7.50, 6.00, and 1.17 rpm. These calculations characterize the kinematic trade-off without establishing real actuator capability or transmission efficiency.

The model inspection also identifies the main barriers to quantitative performance claims: inconsistent load definitions between variants, missing controlled-spur initialization, author-specific CAD paths, and secondary feedback paths that mix translational sensing with angular conversion. Resolving these issues and exporting response data would convert the present design and modeling study into a reproducible control-performance comparison. The current evidence supports the transmission calculations and documented model structure; it does not establish a best-performing closed-loop design.

Appendix A. Code and Results for Publication

The publication folder contains code/, report/, and results/. Model, script, geometry, and library paths in Table A1 are relative to code/; video and report paths are relative to the publication root. Descriptive names distinguish the principal variants. Packaged source files retain their original contents.

Original item	Proposed repository path
design1_matlab/gearbox1_multibody.slx	models/spur_beam_baseline.slx
2gears.zip → 2gear.slx	models/spur_ball_beam_pid.slx
design3.slx	models/worm_ball_beam_pid.slx
design1_matlab/number1.m	scripts/initialize_spur_parameters.m
design1_gearboxes/*.stl	geometry/spur_20_80/ [see upload guide]
2gears.zip → *.stl	geometry/spur_10_50/ [see upload guide]
worm.zip → helical STL and STEP	geometry/worm/ [see upload guide]
Libraries.zip → Libraries/	libraries/Multibody_Parts/ [runtime files]
design1_video.mp4	results/spur_gear_animation.mp4
design2_video.mp4	results/compound_gear_animation.mp4
New English report	report/gear_transmission_ball_beam_report.pdf

Table A1. Core publication files and proposed names. The separate upload guide lists every geometry asset.

A.1 Effective baseline initialization

The following listing reproduces the effective parameter assignments in the baseline script. The module is expressed in centimeters because the relevant saved transforms and constraint radii use centimeters. This initialization belongs to the 20/80-tooth baseline, not the 10/50-tooth controlled model.

```
m = 0.2;            % module [cm]
z1 = 20; z2 = 80;
r1 = m*z1/2;        % pitch radius [cm]
r2 = m*z2/2;
l = 0.5; l1 = 6;
offset = 360/z2/2;  % half-tooth angle [deg]
```

A.2 Publication and execution notes

Omit generated Simulink cache folders, .slxc files, and duplicate archives. Keep the older release model and the negative-command worm variant only if their historical or comparative role is explained. Retain the supporting library code and image files used by its masks; its illustration slide decks are optional. The file-renaming guide records the packaged geometry names and supporting library contents.

Renaming or copying a model does not repair its stored CAD paths or supply missing workspace variables. The 10/50-tooth geometry requires radii of 1.5 cm and 7.5 cm; its remaining linkage variables must be checked against the intended assembly. Model names, paths, signal units, and initialization should be verified before the repository is described as executable without modification.